

Three Rigidity Properties of Logarithmic Weight Spectra for Cocompact Lattice Actions on $\mathbb{H}^3$

Marvin L. Gentry¹

¹Independent Researcher, drmlgentry@protonmail.com, ORCID:
0009-0006-4550-2663

Abstract

Let Γ be a torsion-free cocompact lattice in $\mathrm{PSL}(2, \mathbb{C})$, acting on hyperbolic 3-space $\mathbb{H}^3$ by orientation-preserving isometries. Fix a basepoint $0 \in \mathbb{H}^3$ and a positive constant $\kappa > 0$, and define the logarithmic weight $q(\gamma) = \kappa d(0, \gamma \cdot 0)$ for $\gamma \in \Gamma$. We prove three structural properties of the resulting weight spectrum:

- (i) **Uniqueness.** Every boundary direction $\xi \in \partial\mathbb{H}^3$ has at most one minimal representative in Γ ; since Γ is torsion-free, this representative is unique whenever it exists.
- (ii) **Discreteness.** The weight spectrum $\{q(\gamma) : \gamma \in \Gamma\}$ has a positive minimum value $\Delta = \kappa \ell_0$, where ℓ_0 is the systole of the action. In particular, for any $R > 0$, only finitely many elements satisfy $q(\gamma) \leq R$.
- (iii) **Coordinate rigidity.** Given a finite collection of boundary directions with pairwise angular separation at least $2\varepsilon_0 > 0$, any deformation of the directions by less than ε_0 preserves both the minimal representative for each direction and its integer coordinates under the Milnor–Švarc quasi-isometry.

All three properties follow from proper discontinuity, cocompactness, torsion-freeness, and the classification of isometries of $\mathbb{H}^3$. No arithmetic or dynamical assumptions are required.

1 Introduction

Discrete logarithmic hierarchies arise naturally in geometric contexts such as warped product geometries, renormalisation group flows, and holographic bulk-boundary correspondences. When such a hierarchy is generated by a cocompact group action on hyperbolic space, it is natural to ask what structural constraints the geometry imposes on the weight spectrum. This note provides three such constraints, each elementary but useful as a foundation for applications.

The setting is a torsion-free cocompact lattice $\Gamma \subset \mathrm{PSL}(2, \mathbb{C})$ and the associated weight function $q(\gamma) = \kappa d(0, \gamma \cdot 0)$. The weight spectrum $\mathcal{W} = \{q(\gamma) : \gamma \in \Gamma\}$ is the object of study. Our main results establish:

- *Uniqueness* (Theorem 3.4): each ideal boundary direction is represented by at most one element of minimal weight.
- *Discreteness* (Lemma 4.2 and Corollary 4.4): the spectrum has a positive minimum value equal to κ times the systole; it accumulates nowhere.
- *Coordinate rigidity* (Theorem 5.4): small deformations of the boundary data leave the discrete structure unchanged.

These results are purely geometric. The proofs use only the cocompactness and torsion-freeness of Γ , together with standard facts about isometries of $\mathbb{H}^3$ [4, 3, 1].

2 Geometric Setting

Throughout, $\mathbb{H}^3$ denotes real hyperbolic 3-space with the standard metric of constant curvature -1 , and $\partial\mathbb{H}^3 \cong \mathbb{S}^2$ its sphere at infinity.

Assumption 2.1. Γ is a torsion-free group acting properly discontinuously and cocompactly on $\mathbb{H}^3$ by orientation-preserving isometries. Fix a basepoint $0 \in \mathbb{H}^3$.

Definition 2.2 (Logarithmic weight). For a positive constant $\kappa > 0$, the *logarithmic weight* of $\gamma \in \Gamma$ is $q(\gamma) = \kappa d(0, \gamma \cdot 0)$. The *weight spectrum* of (Γ, κ) is $\mathcal{W}(\Gamma, \kappa) = \{q(\gamma) : \gamma \in \Gamma\}$.

Definition 2.3 (Isometry types). An isometry $\delta \in \mathrm{Isom}(\mathbb{H}^3)$ is: *elliptic* if it fixes a point of $\mathbb{H}^3$; *parabolic* if it fixes exactly one point of $\partial\mathbb{H}^3$ and no point of $\mathbb{H}^3$; *hyperbolic* (or loxodromic) if it fixes exactly two points of $\partial\mathbb{H}^3$.

Lemma 2.4 (No parabolics in cocompact lattices). *A cocompact lattice $\Gamma \subset \mathrm{Isom}(\mathbb{H}^3)$ contains no parabolic elements. Every nontrivial $\gamma \in \Gamma$ is hyperbolic.*

Proof. See [4, Corollary 12.3.5]: the limit set of a cocompact group is all of $\partial\mathbb{H}^3$, so the quotient $\Gamma \backslash \mathbb{H}^3$ is compact. A parabolic element fixes a point on $\partial\mathbb{H}^3$, and that fixed point would be a cusp point of the quotient, contradicting compactness. $\square$

3 Uniqueness of Minimal Representatives

Definition 3.1. We say $\gamma \in \Gamma$ *converges to* $\xi \in \partial\mathbb{H}^3$ if the geodesic ray from 0 in the direction of $\gamma \cdot 0$ has ideal endpoint ξ .

Definition 3.2 (Minimal representative). $\gamma \in \Gamma$ is a *minimal representative* for ξ if it converges to ξ and $q(\gamma) = \inf\{q(\gamma') : \gamma' \in \Gamma, \gamma' \text{ converges to } \xi\}$.

Remark 3.3. Existence of a minimal representative for each ξ follows from cocompactness: since the orbit $\Gamma \cdot 0$ is dense in $\mathbb{H}^3 \cup \partial\mathbb{H}^3$ (the full compactification) and the weight q is proper (finitely many elements of weight $\leq R$ for each R), the infimum is attained.

Theorem 3.4 (Uniqueness). *For every $\xi \in \partial\mathbb{H}^3$, there is at most one minimal representative in Γ .*

Proof. Suppose $\gamma_1, \gamma_2 \in \Gamma$ are two minimal representatives for ξ . Set $\delta = \gamma_1^{-1}\gamma_2$. Since both γ_i converge to ξ , the element δ fixes $\xi \in \partial\mathbb{H}^3$. By Definition 2.3, an isometry fixing a boundary point is either elliptic or parabolic. By Lemma 2.4, Γ contains no parabolics, so δ is elliptic. Since Γ is torsion-free (Assumption 2.1), the only elliptic element is the identity. Hence $\delta = \text{id}$ and $\gamma_1 = \gamma_2$. $\square$

4 Discreteness of the Weight Spectrum

Definition 4.1 (Systole). The *systole* of the action is $\ell_0 = \min_{\gamma \neq \text{id}} d(0, \gamma \cdot 0) > 0$.

The systole is positive because the orbit $\Gamma \cdot 0$ is discrete (proper discontinuity) and Γ has no fixed points in $\mathbb{H}^3$ (torsion-freeness).

Lemma 4.2 (Positive minimum weight). *The weight spectrum has positive minimum value $\Delta := \kappa \ell_0 > 0$. That is, $q(\gamma) \geq \Delta$ for all $\gamma \neq \text{id}$.*

Proof. For $\gamma \neq \text{id}$, we have $d(0, \gamma \cdot 0) \geq \ell_0$ by definition of the systole. Hence $q(\gamma) = \kappa d(0, \gamma \cdot 0) \geq \kappa \ell_0 = \Delta$. $\square$

Remark 4.3. The weight spectrum can have coincidences: distinct elements γ, γ' may satisfy $q(\gamma) = q(\gamma')$ (for instance, γ and any of its conjugates, or γ and γ^{-1} , have the same displacement from 0 if 0 lies on the axis of γ). Lemma 4.2 asserts only that the smallest nonzero weight is Δ , not that all weights are distinct.

Corollary 4.4 (Uniform discreteness). *For any $R > 0$, the set $\{\gamma \in \Gamma : q(\gamma) \leq R\}$ is finite. The weight spectrum $\mathcal{W}(\Gamma, \kappa)$ has no accumulation points.*

Proof. $\{q(\gamma) \leq R\} = \{\gamma : d(0, \gamma \cdot 0) \leq R/\kappa\}$, which is the intersection of Γ with a closed ball of radius R/κ in $\mathbb{H}^3$. This intersection is finite by proper discontinuity. $\square$

5 Coordinate Rigidity under Deformations

Since Γ acts cocompactly on $\mathbb{H}^3$, it is finitely generated. Fix a finite generating set $\mathcal{S}$ and the associated word metric $d_{\mathcal{S}}$ on Γ .

Proposition 5.1 (Milnor–Švarc quasi-isometry [1]). *The orbit map $\phi: (\Gamma, d_{\mathcal{S}}) \rightarrow (\mathbb{H}^3, d)$, $\gamma \mapsto \gamma \cdot 0$, is a quasi-isometry: there exist constants $C_1, C_2, C_3 > 0$ such that*

$$C_1 d_{\mathcal{S}}(\gamma, \gamma') - C_3 \leq d(\gamma \cdot 0, \gamma' \cdot 0) \leq C_2 d_{\mathcal{S}}(\gamma, \gamma') + C_3$$

for all $\gamma, \gamma' \in \Gamma$.

Definition 5.2 (Integer coordinates). For $\gamma \in \Gamma$ with reduced word $w = s_{i_1}^{\epsilon_1} \cdots s_{i_k}^{\epsilon_k}$ ($\epsilon_j = \pm 1$), the *integer coordinate* $a_i(\gamma) = \sum_{j: i_j=i} \epsilon_j$ records the net count of generator s_i . The vector $(a_1, \dots, a_r) \in \mathbb{Z}^r$ depends on the chosen word but is uniquely determined by γ for a fixed generating set and normal form.

Now fix a finite collection of boundary directions $\xi_1, \dots, \xi_m \in \partial\mathbb{H}^3$ with pairwise angular separations

$$\varepsilon_0 := \frac{1}{2} \min_{i \neq j} d_{\mathbb{S}^2}(\xi_i, \xi_j) > 0.$$

Definition 5.3 (Bounded deformation). A *deformation of magnitude ε* is a collection of maps $\xi_i \mapsto \xi'_i$ with $d_{\mathbb{S}^2}(\xi_i, \xi'_i) \leq \varepsilon$ for all i . If $\varepsilon < \varepsilon_0$, the perturbed directions $\xi'_1, \dots, \xi'_m$ remain pairwise distinct, and each ξ'_i lies strictly closer to ξ_i than to any other ξ_j ($j \neq i$).

Theorem 5.4 (Coordinate rigidity). *Let γ_i be the (unique) minimal representative for ξ_i , for each i . For any deformation with $\varepsilon < \varepsilon_0$, the minimal representative for ξ'_i is again γ_i , and its integer coordinates are unchanged.*

Proof. Fix i . For any $\gamma \in \Gamma$ converging to ξ_j with $j \neq i$, the ideal endpoint of the ray from 0 through $\gamma \cdot 0$ lies at angular distance at least $2\varepsilon_0$ from ξ_i , hence at angular distance at least ε_0 from ξ'_i . In contrast, γ_i converges to ξ_i , and its ideal endpoint lies at distance $\leq \varepsilon < \varepsilon_0$ from ξ'_i . Hence γ_i still converges to ξ'_i (in the sense of Definition 3.1) and no element converging to any ξ'_j , $j \neq i$, can converge to ξ'_i . Since the orbit is discrete and the weight is proper (Corollary 4.4), the minimal weight for ξ'_i is attained at γ_i . By Theorem 3.4, γ_i is the unique minimal representative for ξ'_i , and its integer coordinates are the same. $\square$

6 Conclusion

We have established three rigidity properties of the logarithmic weight spectrum of any torsion-free cocompact lattice acting on $\mathbb{H}^3$: uniqueness of minimal representatives, existence of a positive minimum weight, and stability of the discrete structure under bounded deformations of the boundary data. The proofs are elementary and self-contained; they require no special arithmetic structure and apply equally to arithmetic and non-arithmetic lattices.

These properties provide a rigorous geometric foundation for applications in which discrete logarithmic hierarchies arise from cocompact hyperbolic group actions, such as the derivation of discrete weight spectra in symmetric space constructions [2].

References

- [1] M. R. Bridson and A. Haefliger. *Metric Spaces of Non-Positive Curvature*. Springer, 1999.
- [2] M. L. Gentry. Geometric unification of flavor: Masses, mixing, and CP from the golden ratio lattice. *Preprint*, 2026. arXiv preprint (in preparation).
- [3] B. Maskit. *Kleinian Groups*. Springer, 1988.
- [4] J. G. Ratcliffe. *Foundations of Hyperbolic Manifolds*. Springer, 3rd edition, 2019.